

(Students are allowed to use paper materials. Answer directly on the exam paper.)

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BÀI THI GỒM 30 CÂU HỎI (CÓ 20 CÂU 0.25 POINT VÀ 10 CÂU 0.5 POINT)

Score: 28/30 →
8.0

Project: 9.0

Final Score:
 $(8.0 + 9.0) / 2 =$
8.5

Câu 1	Câu 2	Câu 3	Câu 4	Câu 5	Câu 6
				A	C
Câu 7	Câu 8	Câu 9	Câu 10	Câu 11	Câu 12
A	A			A	E
Câu 13	Câu 14	Câu 15	Câu 16	Câu 17	Câu 18
C	B	C	D	B	F
Câu 19	Câu 20	Câu 21	Câu 22	Câu 23	Câu 24
C	A	D	F	C	E
Câu 25	Câu 26	Câu 27	Câu 28	Câu 29	Câu 30
E	D	B			E

FOR MULTIPLE-CHOICE, STUDENTS SHOULD SELECT 01 CORRECT ANSWER AND FILL IN THE ANSWER SHEET.

Question 1 – 5 use the following information. Students need to write out the formula, substitute the numbers, and calculate the final answer in percentage (%) for the questions 1 – 4. Round to the nearest hundredth, for example: 12.34%.

We have a model M that supports diagnosing a type of disease X based on images of damaged skin. We evaluate this model on a dataset of 100 correctly labeled images and obtain the following results:

		Dự đoán	
		Positive	Negative
Nhãn đúng	Positive	45	5
	Negative	2	48

Question 1. Accuracy is defined as the ratio of correctly predicted cases to the total number of cases. Calculate the accuracy of model M. (G1, G2)

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+FN+TN} = \frac{45+48}{45+2+5+48} = 93\%$$

Question 2. Precision is defined as the ratio of correctly predicted positive cases to the total number of cases predicted as positive. Calculate the precision of model M. (G1, G2)

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{45}{45+2} = 95,74\%$$

Question 3. Recall is defined as the ratio of correctly predicted positive cases to the total number of positive samples. Calculate the recall of model M. (G1, G2)

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{45}{45+5} = 90\%$$

Question 4. (0.5 point) Given that the probability of any randomly chosen person from the population having disease X is 0.01%, when a person P is predicted by model M to have disease X (positive), what is the probability that person P actually has disease X? (G1, G2)

$$\frac{45}{47} * 0.01\% = 0.00957\%$$

.....

Question 5. (0.5 point) It is known that the probability of any randomly selected person from the population having disease X is 0.01%. Is model M an effective model to use for tracking disease X in the population? (G1, G2)

A.	Yes. Because M has very high accuracy.
B.	No. Because 98.51% of positive cases are actually not sick.
C.	No. Because 99.77% of positive cases are actually not sick.
D.	No. Because 99.91% of positive cases are actually not sick.

Question 6. We use an iterative method to find the optimal values for the vector of parameters $\mathbf{w}$ of a machine learning model, aim to minimize the loss function $L(\mathbf{w})$. At each iteration t , the value of $\mathbf{w}$ is updated as follows:

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \eta_t \mathbf{d}_t$$

with $\eta_t > 0$ is the learning rate.

What is the formula of $\mathbf{d}_t$ that we should choose? (G2)

A. $\mathbf{d}_t = \nabla L(\mathbf{w}_t)$	B. $\mathbf{d}_t = \mathbf{v}_t$ with $\mathbf{v}_t^T \nabla L(\mathbf{w}_t) = 0$	C. $\mathbf{d}_t = \mathbf{v}_t$ with $\mathbf{v}_t^T \nabla L(\mathbf{w}_t) < 0$	D. $\mathbf{d}_t = \mathbf{v}_t$ with $\mathbf{v}_t^T \nabla L(\mathbf{w}_t) > 0$
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Question 7. We have $f(\mathbf{x}) = 5x_1x_2 + x_3^2$ is a mapping $\mathbb{R}^3 \rightarrow \mathbb{R}$. What is the relation between the vector $\mathbf{d} = (-2, 0, 1)$ and $f(\mathbf{x})$ at the point $\mathbf{x} = (0, 1, 2)$? (GI)

A.	$\mathbf{d}$ is an ascent direction of $f(\mathbf{x})$ at $\mathbf{x} = (0, 1, 2)$.
B.	$\mathbf{d}$ is a descent direction of $f(\mathbf{x})$ at $\mathbf{x} = (0, 1, 2)$.
C.	$\mathbf{d}$ is a gradient vector of $f(\mathbf{x})$ at $\mathbf{x} = (0, 1, 2)$.
D.	$\mathbf{d}$ has the same direction as the contour line of $f(\mathbf{x})$ at $\mathbf{x} = (0, 1, 2)$.

Question 8. If the Hessian matrix $H_f(\mathbf{x})$ of a function $f(\mathbf{w})$ exists at point $\mathbf{w} = \mathbf{x}$, what we can conclude about $H_f(\mathbf{x})$? (GI)

A.	$H_f(\mathbf{x})$ is a positive definite matrix.
B.	$H_f(\mathbf{x})$ is a semi-positive definite matrix.
C.	$H_f(\mathbf{x})$ is an indefinite matrix.
D.	$H_f(\mathbf{x})$ is a symmetric matrix.

Question 9. We have x, y, z corresponding to the *last 3 digits of your student ID*. Find the orthogonal projection of vector $\mathbf{b} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ onto the column space of the matrix $Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$?

We have $\mathbf{b} = \begin{bmatrix} 7 \\ 1 \\ 6 \end{bmatrix}$. The orthogonal projection of $\mathbf{b}$ onto Q :

$$\mathbf{b}' = Q(Q^T Q)^{-1} Q^T \mathbf{b} = \begin{bmatrix} 7 \\ 0 \\ 6 \end{bmatrix}$$

Question 10.(0.5 point) We have x, y corresponding the *last 2 digits of your student ID*.

Find the orthogonal projection of vector $\mathbf{b} = \begin{bmatrix} x \\ y \end{bmatrix}$ onto vector $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$? (GI)

We have $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. The orthogonal projection of $\mathbf{b}$ onto $\mathbf{u}$:

$$\mathbf{b}' = \frac{\mathbf{b} \cdot \mathbf{a}}{\|\mathbf{b}\|^2} \cdot \mathbf{b} = \begin{bmatrix} \frac{16}{36} \\ \frac{42}{37} \end{bmatrix}$$

Question 11 and Question 12 use the following information:

D is the training set consisting of N samples $(\mathbf{x}_i, y_i)$, where $\mathbf{x}_i$ is the feature vector and $y_i \in \mathbb{R}$ is the target value of the training sample i^{th} . The cost function for the Linear Regression model with the vector of parameters $\mathbf{w}$ is defined as:

$$J(\mathbf{w}) = \frac{1}{2N} \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2 = \frac{1}{2N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 = \frac{1}{2N} (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y})$$

We have a matrix X in which each row corresponds to an input feature vector and $\mathbf{y}$ is a vector that contains all N target values.

$$X = \begin{bmatrix} - & - & x_1^T & - \\ - & - & x_2^T & - \\ & & \vdots & \\ - & - & x_N^T & - \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}$$

Question 11. With the provided information, how to compute the vector gradient $\nabla J(\mathbf{w})$? (G1, G2)

A.	$\frac{1}{N} (X^T X \mathbf{w} - X^T \mathbf{y})$
B.	$\frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i) \mathbf{x}_i$
C.	$-\frac{1}{N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i) \mathbf{x}_i$
D.	All are correct.

Question 12. With the provided information, we use Gradient Descent with the learning rate $\eta > 0$ to find the optimal value for the vector of parameters $\mathbf{w}$. What is the rule for updating $\mathbf{w}$? (G1, G2)

A.	$\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial J}{\partial \mathbf{w}}$
B.	$\mathbf{w} \leftarrow \mathbf{w} + \eta \frac{\partial J}{\partial \mathbf{w}}$
C.	$\mathbf{w} \leftarrow \mathbf{w} - \frac{\eta}{N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i) \mathbf{x}_i$
D.	$\mathbf{w} \leftarrow \mathbf{w} + \frac{\eta}{N} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i) \mathbf{x}_i$
E.	A and C are correct.
F.	B and D are correct.

Question 13. Which formula returns the optimal values of parameters for the Linear regression model with L2 regularization? The row i^{th} of the matrix X contains the features of

the datapoint i^{th} in the dataset, and $\mathbf{y}$ is the vector containing all target values $\in \mathbb{R}$ in the dataset. λ is the regularization coefficient. $\mathbf{I}$ is the identity matrix (G1, G2).

A. $(X^T X - \lambda \mathbf{I})^{-1} X^T \mathbf{y}$	B. $\lambda (X^T X + \mathbf{I})^{-1} X^T \mathbf{y}$
C. $(X^T X + \lambda \mathbf{I})^{-1} X^T \mathbf{y}$	D. $-\lambda (X^T X + \mathbf{I})^{-1} X^T \mathbf{y}$

Question 14. Which formula represents the Maximum a posteriori estimation (MAP for the vector of parameters $\mathbf{w}$? D is the dataset of observations. (G1)

A.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmax}} p(D \mathbf{w})$
B.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} D)$
C.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmax}} [\log p(D \mathbf{w}) + \log p(\mathbf{w})]$
D.	A and B are correct.
E.	B and C are correct.
F.	A, B, C are correct.

Question 15. Assume that Y is a random variable corresponding to tossing a coin. $Y=1$ corresponds to getting heads, and $Y=0$ corresponds to getting tails. This is the Bernoulli distribution with the parameter θ that presents the probability of getting heads. Which is the maximum likelihood estimation (MLE) of θ with $N = N_o + N_1$ tossing times? N_1 is the number of times that we get heads, and N_o is the number of times that we get tails. (G1)

A.	$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} -[N_1 \log \theta + N_o \log(1 - \theta)]$
B.	$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \prod_{i=1}^N \theta^{I(y_i=1)} (1 - \theta)^{I(y_i=0)}$
C.	$\hat{\theta} = \frac{N_1}{N_o + N_1}$
D.	$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \sum_{i=1}^N \theta^{I(y_i=1)} (1 - \theta)^{I(y_i=0)}$
E.	All are correct.

Question 16. Which formula represents the training process of the Ridge regression model? λ is the regularization coefficient, N is the number of training samples $(\mathbf{x}_i, y_i)$ in which $\mathbf{x}_i$ is the feature vector and y_i is the target value of the training sample i^{th} . (G1, G2)

A.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2 \right) - \lambda \ \mathbf{w}\ ^2$
B.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2 \right) + \lambda \ \mathbf{w}\ ^2$
C.	$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \left(\frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{w}^T \mathbf{x}_i)^2 \right) + \lambda \ \mathbf{w}\ ^2$
D.	B and C correct.
E.	A, B, and C correct.

Question 17.(0.5 point) Find the suitable values for the parameters of the Linear regression model with the training dataset that has the size of N (N is *relatively small*).

- $\hat{\mathbf{w}}_{OLS}$ is the parameter set found by the Ordinary Least Squares method. This method minimizes the mean squared error (MSE) function.
- $\hat{\mathbf{w}}_{Reg}$ is the parameter set found by minizing MSE and using L2 regularization.

In the following statements, which statement is correct? (G2)

A.	$\hat{\mathbf{w}}_{Reg}$ has a beter performance than $\hat{\mathbf{w}}_{OLS}$ on the training dataset, and $\hat{\mathbf{w}}_{OLS}$ has a beter performance than $\hat{\mathbf{w}}_{Reg}$ on the test dataset.
B.	$\hat{\mathbf{w}}_{OLS}$ has a beter performance than $\hat{\mathbf{w}}_{Reg}$ on the training dataset, and $\hat{\mathbf{w}}_{Reg}$ has a beter performance than $\hat{\mathbf{w}}_{OLS}$ on the test dataset.
C.	$\hat{\mathbf{w}}_{OLS}$ has a beter performance than $\hat{\mathbf{w}}_{Reg}$ on both the training and test datasets.
D.	$\hat{\mathbf{w}}_{Reg}$ has a beter performance than $\hat{\mathbf{w}}_{OLS}$ on both the training and test datasets.

Question 18. Train a Logistic Regression model on a training dataset consisting of N samples. We have $y_i \in \{0, +1\}$ as the true label and $a_i \in \mathbb{R}$ as the predicted probability of the model for the datapoint i^{th} to belong to the class +1. How do we define the loss function L for the whole dataset? (G1, G3)

A.	$L = - \sum_{i=1}^N [y_i \log(a_i) + (1 - y_i) \log(1 - a_i)]$
B.	$L = - \log \left(\prod_{i=1}^N y_i^{a_i} (1 - y_i)^{1-a_i} \right)$
C.	$L = - \log \left(\prod_{i=1}^N a_i^{y_i} (1 - a_i)^{1-y_i} \right)$
D.	$L = \frac{1}{N} \sum_{i=1}^N (y_i - a_i)^2$

E.	A and B are correct.
F.	A and C are correct.
G.	B and C are correct.
H.	A, B, C are correct.

Question 19. Train a Softmax Regression model on a training dataset consisting of N samples for a classification problem having C classes (C is the number of classes). The output of this model $\mathbf{a} = (a_1, a_1, \dots, a_c) \in [0, 1]^C$ is a probability distribution predicted by the model in which a_k is the predicted probability for a sample to belong to the class k . For each datapoint, we have $\mathbf{y} = (y_1, y_1, \dots, y_c) \in \{0, 1\}^C$ is the one-hot vector of the true label of the considering datapoint. How do we define the loss function L for the whole dataset? (G1, G3)

A.	$L = \sum_{i=1}^N \sum_{k=1}^C (y_k^{(i)} - a_k^{(i)})^2$
B.	$L = \sum_{i=1}^N \sum_{k=1}^C y_k^{(i)} \log(a_k^{(i)})$
C.	$L = \sum_{i=1}^N \sum_{k=1}^C -y_k^{(i)} \log(a_k^{(i)})$
D.	$L = \frac{1}{2N} \sum_{i=1}^N \sum_{k=1}^C (y_k^{(i)} - a_k^{(i)})$

Question 20. We build a Softmax Regression model to classify images in the Oxford 17 Flowers dataset. Each image has a size of 224 x 224 and 3 color channels. Each image is assigned 1 of 17 labels (corresponding to 17 different flower species). What is the total number of parameters of the Softmax Regression model? (G1)

A. 2558993	B. 2558976	C. 853009	D. 852992
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Question 21.(0.5 point) We build a neural network to classify images in the Fashion-MNIST dataset. Each image is a 28 x 28 grayscale image. Each image is assigned 1 of 10 labels (t-shirt, trouser, pullover, dress, coat, sandal, shirt, sneaker, bag, root). The network has 2 hidden layers: the first layer has 64 neurons and the second layer has 32 neurons. **Note that each neuron in hidden layers and the output layer has a bias coefficient.** What is the total number of parameters of this neural network? (G1)

A. 153002	B. 52544	C. 152896	D. 52650
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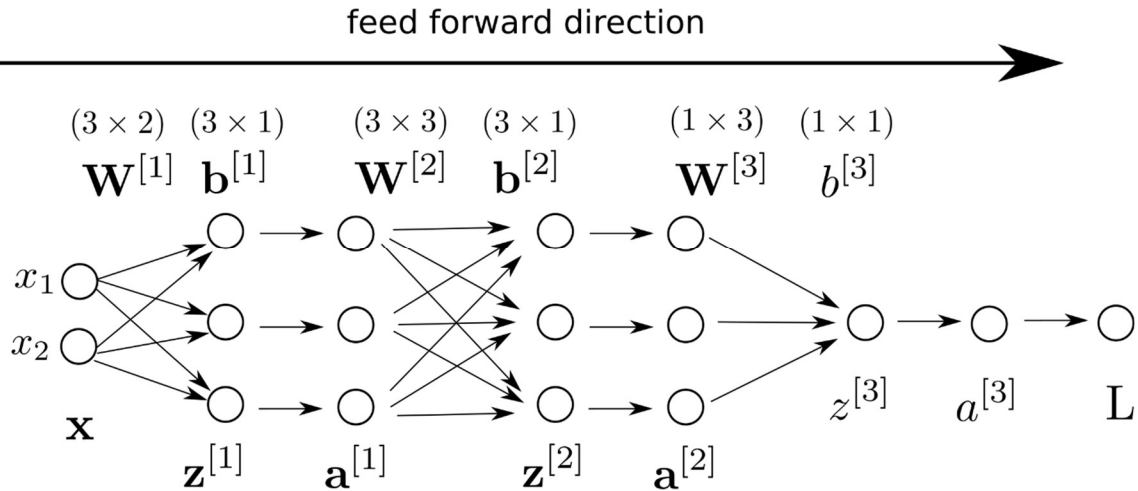
Question 22.(0.5 point) Complex machine learning models can be overfitted if the dataset is not large enough. However, in practice, we do not have information about the true function

$f(\mathbf{x})$, so we do not know in advance what complexity of the machine learning model is appropriate. How do we solve this problem? (G2)

- (A) Only use simple machine learning models to avoid overfitting.
- (B) We can use complex models but need to perform regularization methods.
- (C) We can train many models with different complexities and evaluate the models after training on the test dataset to choose the model with the highest accuracy on the test set.
- (D) We can train many models with different complexities and evaluate the models after training on the validation dataset to choose the model with the highest accuracy on the validation set.
- (E) B and C are correct.
- (F) B and D are correct
- (G) A, B, C, and D are correct

Questions 23, 24, 25 use the following information.

Given a neural network as shown in the following diagram:



Each datapoint $(\mathbf{x}, y)$ has $\mathbf{x}$ as the feature vector and $y \in \{0,1\}$ as the true label.

The output of the above neural network is defined as follows:

$$a^{[3]} = g(W^{[3]}(g(W^{[2]}(g(W^{[1]}\mathbf{x} + \mathbf{b}^{[1]}) + \mathbf{b}^{[2]})) + b^{[3]})$$

We have $W^{[k]}$ and $\mathbf{b}^{[k]}$ as the weight matrix and bias vector at the k^{th} layer of the neural network. g is the activation function at each neuron and is the logistic sigmoid function $a = g(z) = \frac{1}{1+e^{-z}}$. The loss function L is the binary cross entropy loss, defined on the i^{th} training sample as follows:

$$L = -[y^{(i)} \log(a^{[3](i)}) + (1 - y^{(i)}) \log(1 - a^{[3](i)})]$$

Question 23. We have $\mathbf{z}^{[3]} = W^{[3]}\mathbf{a}^{[2]} + \mathbf{b}^{[3]}$. How can we compute $\frac{dL}{db^{[3]}}$? (G2)

A. $\mathbf{a}^{[2]}$	B. $(a^{[3]} - y)(\mathbf{a}^{[2]})^T$	C. $(a^{[3]} - y)$	D. 1
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Question 24.(0.5 point) We have $\mathbf{z}^{[1]} = W^{[1]}\mathbf{x} + \mathbf{b}^{[1]}$. How can we compute $\frac{dL}{dW^{[1]}}$? (G2)

A.	$\begin{bmatrix} \frac{dL}{dz_1^{[1]}} \\ \frac{dL}{dz_2^{[1]}} \\ \frac{dL}{dz_3^{[1]}} \end{bmatrix} [0 \quad x_1 \quad x_2]$
B.	$[0 \quad x_1 \quad x_2]$
C.	$\frac{dL}{d\mathbf{z}^{[1]}} \mathbf{x}^T$
D.	$(a - y) \mathbf{x}^T$
E.	C and D correct.
F.	A and C correct.

Question 25. If we remove the activation functions g at the neurons in the 1st and 2nd hidden layers, how will the neural network operate? (G1, G3)

A.	Nothing changes because we don't need to constrain the output values of the neurons in the hidden layers.
B.	The neural network will have a simpler structure but still achieve the same performance as the original neural network.
C.	The neural network will function like a Linear Regression model.
D.	The neural network will function like a Logistic Regression model.
E.	A and D are correct

Question 26.(0.5 point) Which formula corresponds to the derivative of the logistic sigmoid function? (G1)

$$g(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

A.	$g'(z) = g(z)g(-z)$
B.	$g'(z) = g(z)(1 - g(-z))$
C.	$g'(z) = \frac{e^{-z}}{(1 + e^{-z})^2}$
D.	$g'(z) = \frac{e^z}{(1 + e^z)^2}$
E.	All are correct.

Question 27. What is the disadvantage of the logistic sigmoid function when used as the activation function at the neurons in the hidden layers of a neural network? (G1, G2)

$$g(z) = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$

A.	Exploding gradient $ g(z) \gg 0$ when $z \gg 0$ or $z \ll 0$.
B.	Vanishing gradient bị triệt tiêu $ g(z) \approx 0$ when $z \gg 0$ or $z \ll 0$.
C.	Exploding gradient $ g(z) \gg 0$ when $z \approx 0$.
D.	Vanishing gradient $ g(z) \approx 0$ when $z \approx 0$.
E.	It has no derivative at $z = 0$.

Questions 28, 29 use the following Perceptron model:

The Perceptron model is defined as follows:

$$\hat{y} = \Theta(w_1x_1 + w_2x_2 + \dots + w_nx_n + b) = \Theta(\mathbf{w} \cdot \mathbf{x} + b)$$

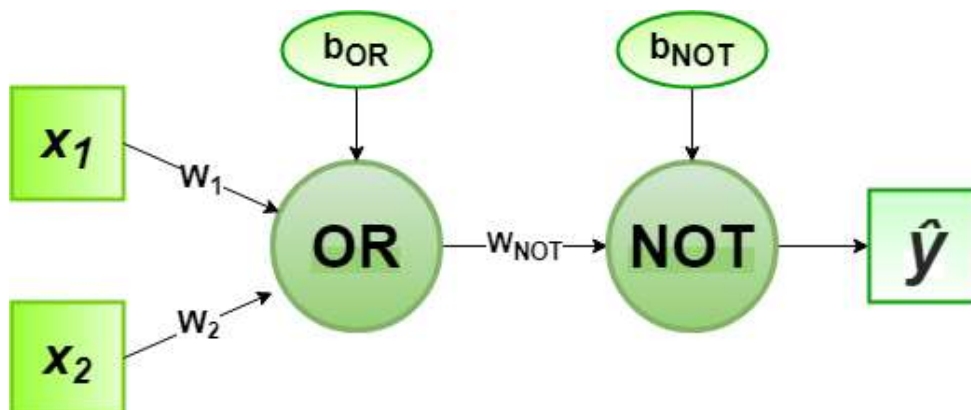
$$\text{with } \Theta(z) = \begin{cases} 1, & v \geq 0 \\ 0, & v < 0 \end{cases}$$

The input features x_i are binary values $x_i \in \{0,1\}$. The weights w and bias b can take real number values $w, b \in \mathbb{R}$.

Question 28.(0.5 point) Truth table of function $y = \mathbf{NOR}(x_1, x_2) = \mathbf{NOT}(\mathbf{OR}(x_1, x_2))$:

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	0

We can use Perceptrons to construct a Multi-Layer Perceptron (MLP) model capable of simulating the truth table of the NOR function. (G3)



With: $\hat{y}_{OR} = \Theta(w_1x_1 + w_2x_2 + b_{OR})$

We have: $\hat{y}_{NOR} = \Theta(w_{NOT}\hat{y}_{OR} + b_{NOT}) = \Theta(w_{NOT}\Theta(w_1x_1 + w_2x_2 + b_{OR}) + b_{NOT})$

Question: Determine the values of $w_{NOT} = ?$, $b_{NOT} = ?$, $w_1 = ?$, $w_2 = ?$, $b_{OR} = ?$ to achieve $\hat{y}_{NOR} = y$.

Hint: Choose values from the set: $\{-2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2\}$.

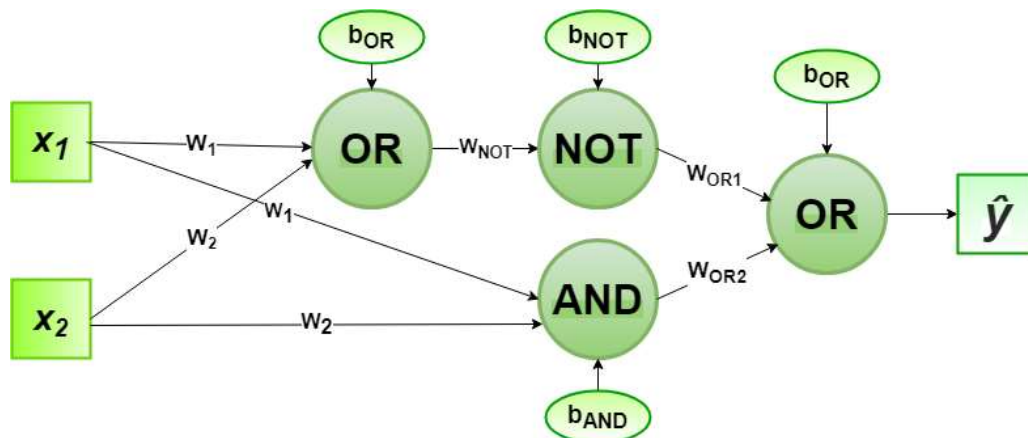
$$w_{NOT} = 1; b_{NOT} = -1; w_2 = 1; b_{OR} = 0; w_1 = 1$$

Question 29.(0.5 point)

Truth table of function $y = \mathbf{XNOR}(x_1, x_2) = \mathbf{OR}(\mathbf{NOT}(\mathbf{OR}(x_1, x_2)), \mathbf{AND}(x_1, x_2))$

x_1	x_2	y
0	0	1
0	1	0
1	0	0
1	1	1

We can use the Perceptrons to construct a Multi-Layer Perceptron (MLP) capable of simulating the truth table of the NOR function. (G3)



$$\hat{y}_1 = \Theta(w_1x_1 + w_2x_2 + b_{OR}), \hat{y}_2 = \Theta(w_1x_1 + w_2x_2 + b_{AND}), \hat{y}_3 = \Theta(w_{NOT}\hat{y}_1 + b_{NOT})$$

$$\text{We have: } \hat{y}_{XNOR} = \Theta(w_{OR1}\hat{y}_3 + w_{OR} \hat{y}_2 + b_{OR})$$

Determine the values of $w_1 = ?$, $w_2 = ?$, $w_{NOT} = ?$, $w_{OR1} = ?$, $w_{OR2} = ?$, $b_{AND} = ?$, $b_{OR} = ?$, $b_{NOT} = ?$ to achieve $\hat{y}_{XNOR} = y$.

Hint: Choose values from the set: $\{-2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2\}$.

Question 30. Which pseudocode corresponds to the steps in the t -th iteration of the Gradient Descent algorithm with Momentum for updating the parameter vector $\mathbf{w}$ to minimize the

objective function $L(\mathbf{w})$? We have α as the learning rate, and β as the momentum coefficient.
(G3)

A.	$\mathbf{g}^{(t)} \leftarrow \nabla_{\mathbf{w}} L(\mathbf{w}^{(t-1)})$ $\mathbf{m}^{(t)} \leftarrow \beta \mathbf{m}^{(t-1)} + (1 - \beta) \mathbf{g}^{(t)}$ $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t)} - \alpha \mathbf{m}^{(t)}$
B.	$\mathbf{g}^{(t)} \leftarrow \nabla_{\mathbf{w}} L(\mathbf{w}^{(t-1)})$ $\mathbf{m}^{(t)} \leftarrow \beta \mathbf{m}^{(t-1)} + (1 - \beta) \mathbf{g}^{(t)}$ $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t)} + \alpha \mathbf{m}^{(t)}$
C.	$\mathbf{g}^{(t)} \leftarrow \nabla_{\mathbf{w}} L(\mathbf{w}^{(t-1)})$ $\mathbf{m}^{(t)} \leftarrow \beta \mathbf{m}^{(t-1)} + (1 - \beta)(-\mathbf{g}^{(t)})$ $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t)} + \alpha \mathbf{m}^{(t)}$
D.	$\mathbf{g}^{(t)} \leftarrow \nabla_{\mathbf{w}} L(\mathbf{w}^{(t-1)})$ $\mathbf{m}^{(t)} \leftarrow \beta \mathbf{m}^{(t-1)} + (1 - \beta)(-\mathbf{g}^{(t)})$ $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t)} - \alpha \mathbf{m}^{(t)}$
E.	A, B are correct.
F.	C, D are correct.

Bảng chuẩn đầu ra môn học CS115:

CĐRMH	Mô tả CĐRMH
G1	Hiểu và trình bày được các cơ sở Toán học trong một số thuật toán Máy học căn bản.
G2	Vận dụng các công cụ Toán học để giải quyết một số bài toán Máy học căn bản
G3	So sánh được các phương pháp Toán và các giải thuật Tin học khi áp dụng giải quyết các vấn đề trong Khoa học máy tính. Trình bày được sự cần thiết của các giải thuật xấp xỉ.